

Experiment- 08

Setting Up a Wireless Network

Lab Objective: To set up and configure a wireless network using a wireless router and secure the network using appropriate security protocols.

Prerequisites: Basic understanding of networking concepts and familiarity with wireless network configuration.

Outcome: Successfully set up and configure a secure wireless network.

Description:

Wireless networks provide a convenient way for devices to communicate and access the internet without the need for physical connections. However, they also introduce security challenges that need to be addressed to protect the network from unauthorized access and data breaches.

Wireless Network Components:

Wireless Router: The central device that connects to the internet and allows wireless devices to communicate with each other and access the internet.

Devices: Computers, smartphones, tablets, and other wireless-enabled devices that connect to the network.

Access Points: Additional devices that extend the wireless coverage of the network.

Wireless Standards: Wireless networks operate based on different standards, which determine the speed and range of the network. The most common standards are:

802.11b: Operates at 2.4 GHz with a maximum speed of 11 Mbps.

802.11g: Operates at 2.4 GHz with a maximum speed of 54 Mbps.

802.11n: Operates at both 2.4 GHz and 5 GHz with a maximum speed of 600 Mbps.

802.11ac: Operates at 5 GHz with a maximum speed of up to several Gbps.

802.11ax (Wi-Fi 6): Operates at both 2.4 GHz and 5 GHz with improved efficiency and speed over previous standards.

Wireless Security Protocols: Securing a wireless network is crucial to prevent unauthorized access and protect data. The main wireless security protocols are:

WEP (Wired Equivalent Privacy):

- o **Description:** The original wireless security standard.
- o **Characteristics:** Uses static encryption keys. Vulnerable to various types of attacks due to weak encryption.
- o **Usage:** Considered obsolete and not recommended for use.

WPA (Wi-Fi Protected Access):

- o **Description:** A security protocol developed to address the weaknesses of WEP.
- o **Characteristics:** Uses TKIP (Temporal Key Integrity Protocol) for dynamic key generation.
- o **Usage:** Provides better security than WEP but has been largely replaced by WPA2.

WPA2 (Wi-Fi Protected Access 2):

- o **Description:** The current widely used wireless security standard.
- o **Characteristics:** Uses AES (Advanced Encryption Standard) for strong encryption. Supports both personal (pre-shared key) and enterprise (RADIUS server) modes.
- o **Usage:** Recommended for most networks due to its strong security features.

WPA3 (Wi-Fi Protected Access 3):

- o **Description:** The latest security protocol designed to provide enhanced security over WPA2.
- o **Characteristics:** Provides better protection against brute-force attacks, improved encryption for open networks, and individualized data encryption.
- o **Usage:** Recommended for new networks were supported by devices.